

Wazuh File Integrity Monitoring Lab

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Purpose: The purpose of this lab is to gain hands-on experience configuring and testing Wazuh's File Integrity Monitoring capabilities. This lab will cover what FIM is and why it is important, how to configure Wazuh to use FIM, and testing and validating that FIM works.

Sections:

1. Introduction to File Integrity Monitoring with Wazuh:
2. Configure File Integrity Monitoring on Agent:
3. Test and Validate:

Section #1 Introduction to File Integrity Monitoring With Wazuh:

1.1 Why is FIM important?

FIM is a vital cybersecurity feature for an organization since it helps monitor the integrity of vital files on systems. For an organization that stores top-secret or mission critical data on devices having the ability to monitor this data is an important capability.

Imagine a company stores vital financial data used in the financial reporting process on an Ubuntu device. The integrity of this data is vital since inaccurate data would cause the financial reports to be an inaccurate representation of the company's finances. The company would most likely face legal repercussions from stockholders and its reputation will be hurt. By having a FIM capability to monitor the integrity of data stored on the Ubuntu system the company would be able to ensure that the financial reports are accurate and since the data has not been tampered with.

FIM is also a requirement for many compliance programs such as PCI DSS, SOX (which would be the compliance requirement for the company mentioned previously), and HIPAA. Failure to implement FIM would mean that a company will face financial and legal issues from regulators.

1.2 What are the key features of Wazuh FIM?

Wazuh's FIM capabilities:

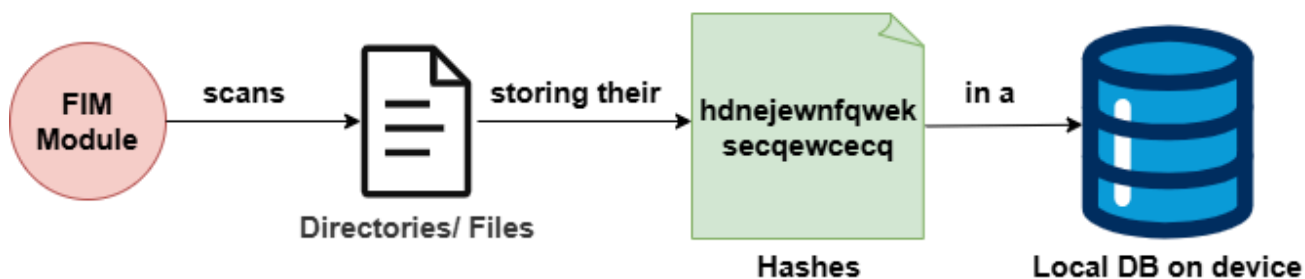
File Monitoring	Scheduled Scans
Directory Monitoring	Custom Rule Creation
Checksum Verification	Change Detection
Real-Time Scans	Detailed Alerts

1. File monitoring: Wazuh FIM can be configured to monitor the integrity of specific files, allowing a more granular approach to FIM.
2. Scheduled Scans: Wazuh allows FIM to perform scheduled scans at regular intervals of your choosing. It can be every few minutes to every 4 hours. It all depends on how often you would like FIM to check the integrity of your data.
3. Directory Monitoring: Wazuh FIM can be configured to check the integrity of all files in a directory. This is useful if you have a certain directory that stores sensitive data on a device or you want to monitor the integrity of an OS configuration directory like /etc or \System32.
4. Custom Rules: FIM allows users to configure custom rules based on a variety of parameters ranging from the specific group a user is a part of to the name of a process that modifies a file.
5. Checksum Verification: Wazuh FIM uses checksums of the monitored files to verify their integrity. FIM supports various checksum algorithms such as SHA-1, SHA-256, and MD5.

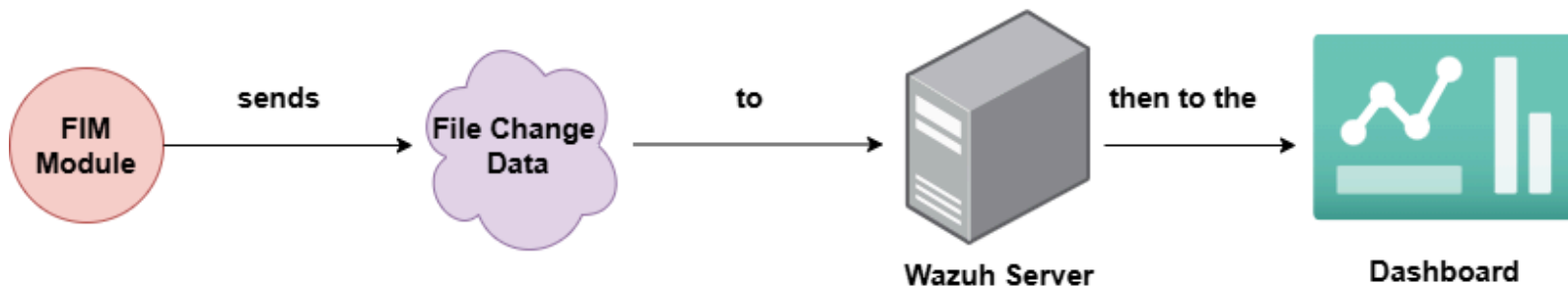
6. Change Detection: FIM can detect if a file is deleted, modified, or has its permissions changed.
7. Real-Time Scans: Wazuh FIM allows you to configure real time monitoring for certain directories. It should be noted that this feature only works with directories and not individual files.
8. Detailed Alerts: FIM can be configured to perform detailed alerts that not only include what happened to a file but what user performed the changes. This is done through the whodata monitoring setting.

1.3 How does Wazuh FIM work?

1. Agent-Side



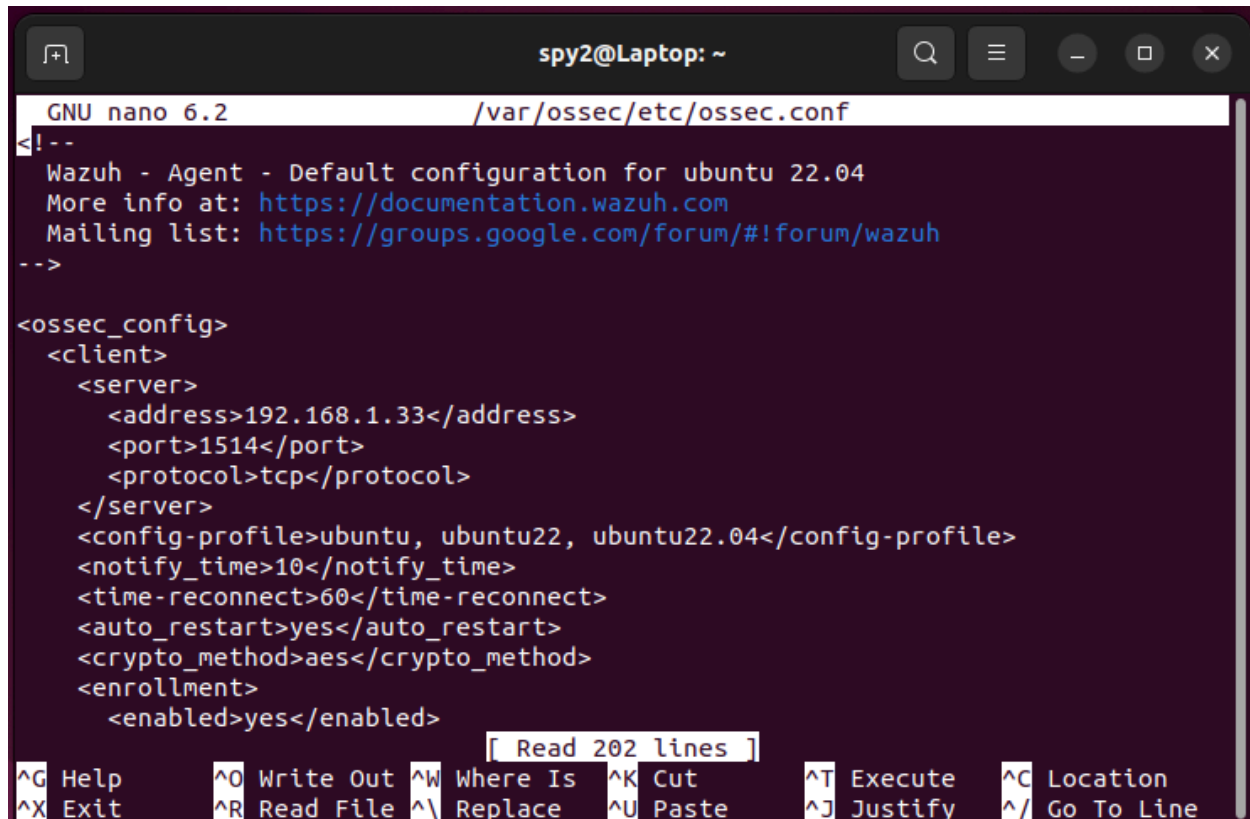
2. Server-Side



Section #2 Configure File Integrity Monitoring on Agent:

2.1 Open Agent configuration file (/var/ossec/etc/ossec.conf):

FIM is configured on the agent-side. The agent-side configuration file is /var/ossec/etc/ossec.conf.



```
GNU nano 6.2 /var/ossec/etc/ossec.conf
<!--
Wazuh - Agent - Default configuration for ubuntu 22.04
More info at: https://documentation.wazuh.com
Mailing list: https://groups.google.com/forum/#!forum/wazuh
-->

<ossec_config>
  <client>
    <server>
      <address>192.168.1.33</address>
      <port>1514</port>
      <protocol>tcp</protocol>
    </server>
    <config-profile>ubuntu, ubuntu22, ubuntu22.04</config-profile>
    <notify_time>10</notify_time>
    <time-reconnect>60</time-reconnect>
    <auto_restart>yes</auto_restart>
    <crypto_method>aes</crypto_method>
    <enrollment>
      <enabled>yes</enabled>
    </enrollment>
  </client>
</ossec_config>
```

[Read 202 lines]

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute	^C Location
^X Exit	^R Read File	^_ Replace	^U Paste	^J Justify	^/ Go To Line

2.2 Enable syscheck:

Wazuh calls FIM Syscheck on the configuration file. The configuration file is also in xml format so be sure to enclose values in `<>` `</>`.

```
GNU nano 6.2 /var/ossec/etc/ossec.conf
<interval>12h</interval>
<skip_nfs>yes</skip_nfs>
</sca>

<!-- File integrity monitoring -->
<syscheck>
  <disabled>no</disabled>

  <!-- Frequency that syscheck is executed default every 12 hours -->
  <frequency>43200</frequency>

  <scan_on_start>yes</scan_on_start>

  <!-- Directories to check (perform all possible verifications) -->
  <directories>/etc,/usr/bin,/usr/sbin</directories>
  <directories>/bin,/sbin,/boot</directories>

  <!-- Files/directories to ignore -->
  <ignore>/etc/mtab</ignore>
  <ignore>/etc/hosts.deny</ignore>

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

Currently Syscheck is enabled in the Ubuntu agent. Syscheck is enabled by default on Wazuh agents so you don't have to enable it manually.

2.3 Define scan frequency:

```
<disabled>no</disabled>

<!-- Frequency that syscheck is executed default every 12 hours -->
<frequency>43200</frequency>

<scan_on_start>yes</scan_on_start>

<!-- Directories to check (perform all possible verifications) -->
<directories>/etc,/usr/bin,/usr/sbin</directories>
<directories>/bin,/sbin,/boot</directories>

<!-- Files/directories to ignore -->
<ignore>/etc/mtab</ignore>
<ignore>/etc/hosts.deny</ignore>
```

The current scan frequency is 12 hours. 43200 is 12 hours in seconds form. I will change this setting to every 4 hours or $60 \times 4 \times 60$ or 14400 seconds. The scan frequency will determine how often the agent scans the files, compares the hashes and other metadata, and sends the data to the server.

```
<!-- Frequency that syscheck is executed default every 12 hours -->
<frequency>14400</frequency>

<scan_on_start>yes</scan_on_start>
```

2.4 Enable initial scan when agent starts:

```
<scan_on_start>yes</scan_on_start>
```

Scan on start is already enabled. Scan on start means that a scan is initially started on the endpoint by the agent when the endpoint is first turned on. The Wazuh agent on my Ubuntu system is enabled via systemctl so it will start when the system boots.

2.5 Specify files and directories you want to check:

The following are the default files and directories checked:

```
<!-- Directories to check (perform all possible verifications) -->
<directories>/etc,/usr/bin,/usr/sbin</directories>
<directories>/bin,/sbin,/boot</directories>
```

I will also add the /tmp directory and a custom directory (/home/spy2/secrets/)

```
<!-- Directories to check (perform all possible verifications) -->
<directories>/etc,/usr/bin,/usr/sbin</directories>
<directories>/bin,/sbin,/boot</directories>
<directories>/tmp,/home/spy2/secrets</directories>
```

Note: The /tmp directory is an interesting directory to watch since files created in it will disappear when the system is rebooted. It is commonly used by attackers to hide evidence from their attacks so monitoring the directory for activity would be a useful incident response ability.

2.6 Enable whodata:

Whodata allows FIM to integrate with system auditing functionalities like auditd to gain a greater view into the actions performed on files and directories. Wazuh can tell the user or the process that modified or accessed a file which is helpful for incident response and auditing purposes.

- First I will install the auditd service onto my Ubuntu VM:

```

spy2@Laptop:~$ sudo apt install auditd
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libauparse0
Suggested packages:
  audispd-plugins
The following NEW packages will be installed:
  auditd libauparse0
0 upgraded, 2 newly installed, 0 to remove and 338 not upgraded.
Need to get 270 kB of archives.
After this operation, 876 kB of additional disk space will be used.
Do you want to continue? [Y/n]

```

- Check if auditd is disabled by default:

```

spy2@Laptop:~$ sudo auditctl -l | grep task
spy2@Laptop:~$

```

- Configure whodata monitoring for certain directories:

I will set whodata monitoring on /etc, /home/spy2/secrets/, and /tmp.

```

<!-- Directories to check (perform all possible verifications) -->
<directories>/usr/bin,/usr/sbin</directories>
<directories>/bin,/sbin,/boot</directories>
<directories whodata="yes">/tmp</directories>
<directories whodata="yes">/etc</directories>
<directories whodata="yes">/home/spy2/secrets</directories>

```

2.7 Save changes and restart agent service:

```

spy2@Laptop:~$ sudo systemctl restart wazuh-agent.service
[sudo] password for spy2:
spy2@Laptop:~$ sudo auditctl -l | grep wazuh
-w /etc -p wa -k wazuh_fim
-w /home/spy2/secrets -p wa -k wazuh_fim
-w /tmp -p wa -k wazuh_fim
spy2@Laptop:~$

```

Section #3 Test and Validate:

3.1 Create test files directories being monitored and write some data to them:

- /home/spy2/secrets/

```
spy2@Laptop:~/secrets$ touch secret1.txt && echo "testing 1..2..3.." > secret1.txt
spy2@Laptop:~/secrets$ cat secret1.txt
testing 1..2..3..
spy2@Laptop:~/secrets$
```

- /etc/

```
spy2@Laptop:~$ cd /etc
spy2@Laptop:/etc$ sudo touch fim_test.txt
spy2@Laptop:/etc$ echo "testing 1 2 3" > fim_test.txt
bash: fim_test.txt: Permission denied
spy2@Laptop:/etc$ sudo echo "testing 1 2 3" > fim_test.txt
bash: fim_test.txt: Permission denied
spy2@Laptop:/etc$ sudo echo "testing 1 2 3" >> fim_test.txt
bash: fim_test.txt: Permission denied
spy2@Laptop:/etc$ sudo -i
root@Laptop:~# cd /etc
root@Laptop:/etc# echo "testing 1 2 3" > fim_test.txt
root@Laptop:/etc# exit
logout
spy2@Laptop:/etc$
```

- /tmp/

```
spy2@Laptop:~$ cd /tmp
spy2@Laptop:/tmp$ touch testfile.txt
spy2@Laptop:/tmp$ echo "testing 1 2 3" > testfile.txt
spy2@Laptop:/tmp$
```

3.2 Delete the test files

- /home/spy2/secrets/secret1.txt

```
spy2@Laptop:~$ cd secrets/  
spy2@Laptop:~/secrets$ rm secret1.txt  
spy2@Laptop:~/secrets$
```

- /etc/fim_test.txt

```
spy2@Laptop:~/secrets$ cd /etc  
spy2@Laptop:/etc$ sudo rm fim_test.txt  
spy2@Laptop:/etc$
```

- /tmp/testfile.txt

```
spy2@Laptop:~$ cd /tmp  
spy2@Laptop:/tmp$ rm testfile.txt  
spy2@Laptop:/tmp$
```

3.3 Verify the FIM alerts are generated in the Wazuh web portal:

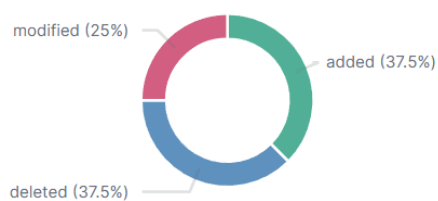
1. Look at the FIM module dashboard:

On the Wazuh FIM dashboard key information such as most active users, actions, and the names of files added, modified, and deleted are displayed in a pie chart format.

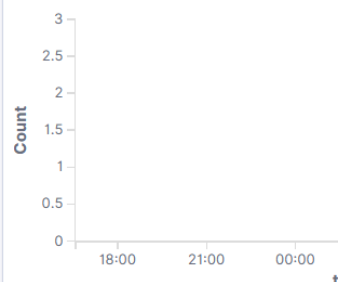
Most active users



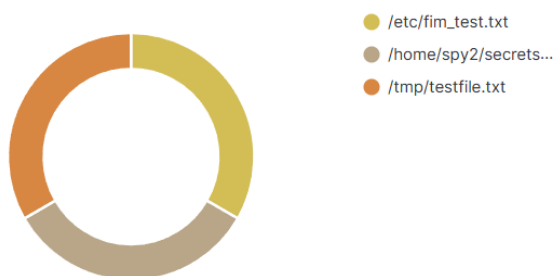
Actions



Events



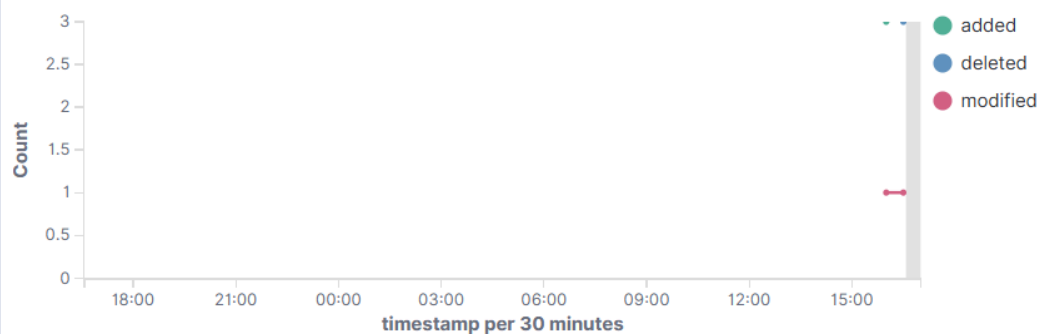
Files added



Files modified



Events



Files deleted

